DeadlockExample.java:->

```
package com.example.demo;

import java.util.concurrent.locks.ReentrantLock;

public class DeadlockExample {
    private static final ReentrantLock lock1=new ReentrantLock();
    private static final ReentrantLock lock2=new ReentrantLock();

    public static void main(String[] args) {
        Runnable task1=()->{
            lock1.lock();
            try{
                System.out.println("Thread-1 acquired lock1. waiting for
lock2 ...");
                Thread.sleep(100L);
                lock2.lock();
                try{
                    System.out.println("Thread-1 acquired lock2.");
                } finally {
                   lock2.unlock();
                }

            } catch (InterruptedException e) {
                Thread.currentThread().interrupt();
            }
            finally {
                lock1.unlock();
            }
        };
Runnable task2=()->{
    lock2.lock();
    try{
        System.out.println("Thread-2 acquired lock2.waiting for lock1 ..." );
        Thread.sleep(100L);
        lock1.lock();

    try {
        System.out.println("Thread-2 acquired lock1.");
    }finally {
        lock1.unlock();
    }
} catch (InterruptedException e) {
        Thread.currentThread().interrupt();
    }finally {
        lock2.unlock();
    }
    };

Thread thread1=new Thread(task1);
Thread thread2=new Thread(task2);
thread1.start();
thread2.start();}
}
```